

Quiz · Ohm's Law for a Closed Circuit

Cells connected in series and the rheostat — five challenging questions. Read every figure carefully before you choose.

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PHYSICS

5 questions

Time: 25 minutes

Mark: 5

Name:

Class:

Mark: / 5

Consider the connecting wires to have negligible resistance, the ammeter to be ideal (negligible resistance) and the voltmeter to be ideal (very large resistance, so it draws no current).

- 1 In the circuit shown, when the ammeter reads 2 A the voltmeter reads 10 V. When the slider of the rheostat is moved, the ammeter reads 4 A and the voltmeter reads 8 V. **The maximum current** that the cell can deliver (when its terminals are joined by a wire of negligible resistance) is:

MCQ

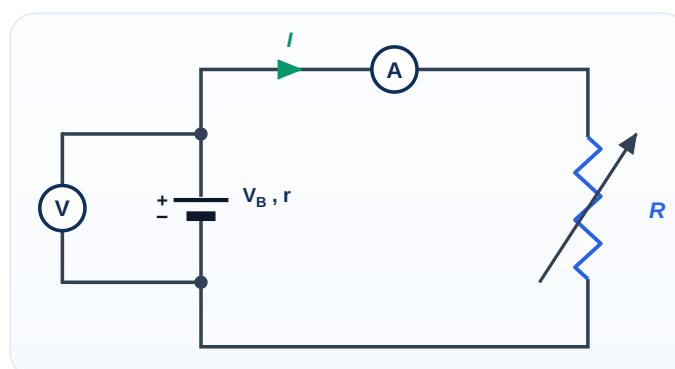


FIGURE (1)

A 5 A

B 6 A

C 12 A

D 24 A

- 2 The terminals of a cell are connected to a $2\ \Omega$ resistor and a current of 3 A flows. When the resistor is replaced by a $5\ \Omega$ resistor, the current becomes 1.5 A. **The emf of the cell and its internal resistance** are respectively:

MCQ

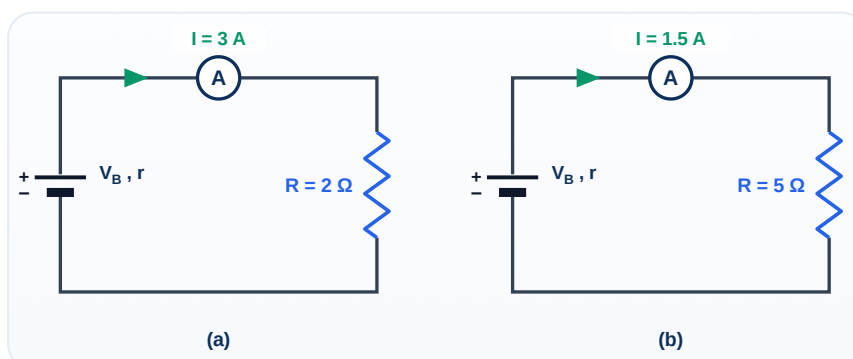


FIGURE (2)

A 9 V, $1\ \Omega$

B 7.5 V, $0.5\ \Omega$

C 12 V, $2\ \Omega$

D 6 V, zero

MCQ

- 3 A battery consists of 6 identical cells connected in series. The emf of each cell is 1.5 V and its internal resistance is $0.5\ \Omega$, and the battery is connected to an external resistance of $6\ \Omega$. If the fourth cell is connected **in reverse** by mistake, as shown in the figure, the ratio of the current **after** the mistake to the current **before** it is:

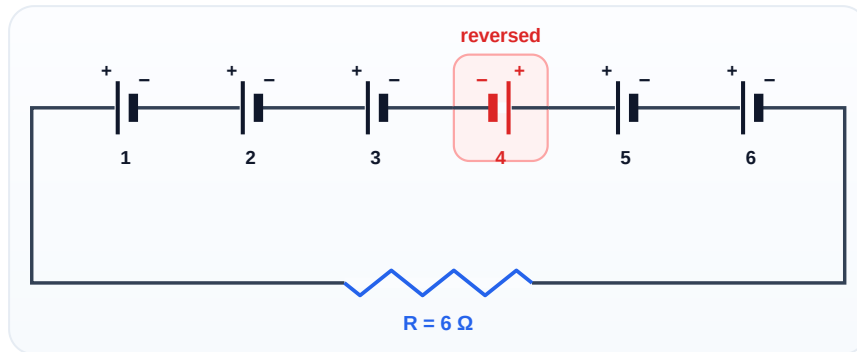


FIGURE (3)

- A $5/6$ B $2/3$ C $3/4$ D $1/2$

- 4 In the circuit shown, two batteries are connected in series **in opposition** (the positive terminal of one is connected to the positive terminal of the other). **The reading of the voltmeter** connected across the terminals of battery (2) is:

MCQ

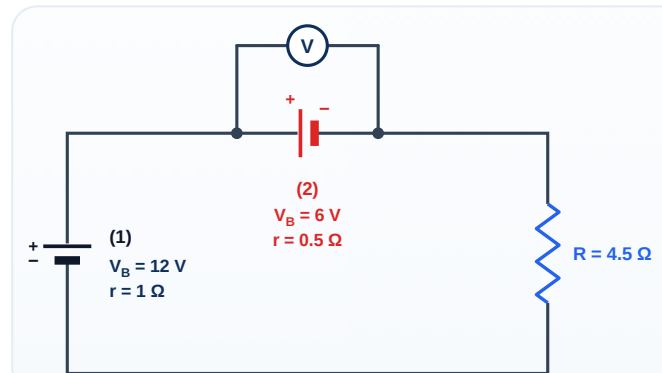


FIGURE (4)

- A 6 V B 5.5 V
C 11 V D 6.5 V

- 5 In the circuit shown, the rheostat is connected through its end (a) and its sliding contact (S), while its end (b) is left free. If the sliding contact S is moved **towards end (a)**, then **the reading of the voltmeter**:

MCQ

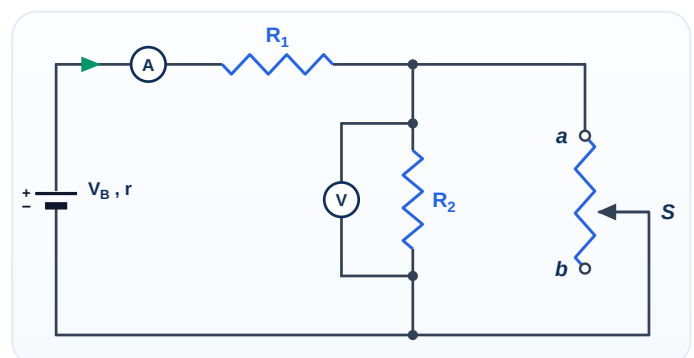


FIGURE (5)

- A increases B decreases C remains constant